Gazi Faysal Jubayer

Mechanical Engineer | CAD & Formula Student | Machine Learning & Full-Stack
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Professional Summary

Mechanical Engineering student at KUET who builds across CAD and software. Vice Captain (Technical) of Team Kilo Flight, leading the design and build of a Formula Student race car. Champion of CAD Conqueror (EELEVEN AutoFest 2026) and 4-time national CAD contest podium winner; CSWA, CSWP, CSWPA-SM and CSWPA-WD certified. 2-time NASA Space Apps Global Nominee and published researcher who ships ML, computer-vision and full-stack products.

Education

Khulna University of Engineering and Technology (KUET) — Khulna, Bangladesh
Bachelor of Technology in Mechanical Engineering — *Dec. 2022 – June 2027*

Khulna Public College — Khulna, Bangladesh
Higher Secondary Certificate (HSC), Science — *2019 – 2021*

Experience

Team Kilo Flight — KUET Formula Student — Khulna, Bangladesh
Vice Captain (Technical) — *Oct. 2025 – Present*

- Lead technical operations across chassis & suspension, body & aerodynamics, powertrain, electrical & DAQ and braking & steering to build **KILOFLIGHT PHOENIX**, a combustion Formula Student car for **FSAE Dynamix India 2027**
- Design and optimize the powertrain in SolidWorks: 292 cc CFMoto 300SR single-cylinder engine integration, chain final drive, fuel and cooling systems; mentor sub-teams on CAD best practices
- Placed **3rd** at the International FS Concept Challenge 2025 (Formula Bharat); previously Senior Executive, Mechanical Powertrain

ONNEXX — Texas, USA (Remote)
Jr. Mechanical Engineer (Part-time) — *Oct. 2026 – Present*

- Mechanical engineering for the Project ONNEXX sound system

3D Bytes — Khulna, Bangladesh
Founder & Technical Lead — *Jan. 2026 – Present*

- Run a commercial 3D printing service for prototyping and custom fabrication on Bambu Lab P1S, tuning Bambu Studio / OrcaSlicer profiles for strong, engineering-grade parts
- Built the studio's Next.js + Supabase platform with an in-browser Gridfinity generator (OpenSCAD WASM, STL export)

Team Durbar — KUET Mars Rover — Khulna, Bangladesh
Autonomous Lead — *Sept. 2025 – Present*

- Direct rover autonomy (perception, path planning, sensor integration) with ROS, OpenCV and Python; **2nd place**, NASA Texas Space Grant Mars Autonomous Rover Rally 2025; member since July 2024

Team BengalSub — RoboSub 2025 — Khulna, Bangladesh
Website Team Lead & Mechanical Designer — *May 2025 – Jan. 2026*

- Led the team website (bengalsub.com) and designed AUV mechanical components in SolidWorks for hydrodynamics and structural integrity

Uddhar (uddharbd.org) — Bangladesh
Co-Founder & Chief Information Officer — *Aug. 2024 – Present*

- Co-founded a disaster-relief platform during the 2024 Feni flood; lead a 30+ member technical team building the Django web platform and Flutter app

VSB — Virtual Solution Book (vsbbd.com) — Remote
Co-Founder & Tech Lead — *Jan. 2024 – Present*

- Co-founded an EdTech platform serving 1000+ HSC and university-admission students; lead Django development with Selenium content automation

LOOP — Khulna, Bangladesh
Project Manager (promoted from Assistant PM, Oct. 2025) — *Sept. 2026 – Present*

CADers, KUET — Khulna, Bangladesh
Chief Policy Analyst — KUET's CAD club, 100+ members — *Jan. 2025 – Present*

Fab Lab KUET Khulna, Bangladesh
Student Operator *July 2026 – Present*

- Operate 3D printers, laser cutters and CNC routers, helping students and researchers turn CAD designs into prototypes

Notable Projects

Mars Autonomous Rover Navigation | *ROS, Python, Space Teams PRO* 2025

- Autonomous navigation for Mars Kaiser Crater in a physics-based simulation: path planning and obstacle avoidance; **2nd place**, NASA Texas Space Grant rally

ExoBengal — AI Exoplanet Detection | *TensorFlow, Scikit-learn, Next.js, FastAPI* 2024

- PyPI package of RF, CNN, KNN and Decision Tree models at 90% accuracy (0.90 AUC-ROC) on NASA Kepler Q1–Q17, with a Next.js site of 3D orbit visualizations and an AI chatbot

Custom TPMS Infill for OrcaSlicer | *C++, OrcaSlicer, Claude AI* 2026

- Reverse-engineered OrcaSlicer's infill code to add equation-driven TPMS-P and TPMS-IWP patterns with an adaptive-density gradient (1m 10s and 0.36 g saved per test bar, slicer estimate)

Mine Shredder Machine | *SOLIDWORKS* 2024

- 21-item belt-driven rotary shredder with fixed and adjustable blades on 6008 bearings; full assembly and exploded drawings, motion study; 180+ GrabCAD downloads

Human Body Blur | *Python, YOLOv8, ByteTrack, OpenCV* 2025

- Anonymizes chosen people in video with YOLOv8 instance masks and ByteTrack IDs, blurring only selected tracks while preserving audio

Bangla Civic Intelligence Suite | *Next.js, Hugging Face, Transformers.js* 2025

- Serverless Bangla platform: OCR + voice document Q&A with spoken answers, and AI civic-hotspot detection on Leaflet heatmaps, using free open-source models

WealthPilot (Team Joker) | *Next.js, Prisma, PostgreSQL* 2026

- Personal-finance workspace for the Shenzhen International FinTech Competition 2026: exact-money ledger, on-device statement/receipt import, grounded AI copilot

Achievements & Awards

Champion — CAD Conqueror, EELEVEN Bangladesh AutoFest 2026 (car concept design & fabrication)
Champion — MIE Robolution CAD Contest, CUET (Team Phenol; 17,000 BDT prize)
4-time national CAD podium — also 1st Runner-Up at Calibration 1.0 (Feb. 2024), Calibration 2.0 (May 2026) and Energy Fest 1.0
3rd Place — International FS Concept Challenge 2025, Formula Bharat (Team Kilo Flight)
2nd Place — NASA Texas Space Grant Mars Autonomous Rover Rally 2025 (Team Durbar)
2× Galactic Problem Solver & Global Nominee — NASA Space Apps Challenge 2023 (Regional 1st Runner-Up) and 2024 (Regional 2nd Runner-Up)
RoboSub 2025 Competitor — Team BengalSub, autonomous underwater vehicle

Technical Skills

CAD & Fabrication: SolidWorks (CSWP, CSWPA-SM, CSWPA-WD), AutoCAD, Ansys, Blender, KeyShot, 3D printing, Sheet Metal & Weldments
Programming: Python, C++, C, TypeScript, JavaScript, Dart, SQL
ML & Robotics: TensorFlow, Scikit-learn, PyTorch, OpenCV, YOLOv8, ROS, Pandas, NumPy
Web & Apps: Next.js, React, Django, FastAPI, Flask, Flutter, Tailwind CSS, Supabase, PostgreSQL, Docker

Certifications

Dassault Systèmes: CSWPA-WD (Weldments), CSWPA-SM (Sheet Metal), CSWP (Mechanical Design), CSWA — April–July 2024
IBM: Python Project for AI & Application Development; Developing AI Applications with Python & Flask; Python for Data Science, AI & Development (2024)

Publications

M. A. Alam, M. Rafiquzzaman, M. H. Ali, **G. F. Jubayer**, "Analyzing the impact of opportunistic maintenance optimization on manufacturing industries in Bangladesh: An empirical study," *BenchCouncil Transactions on Benchmarks, Standards and Evaluations*, vol. 4, no. 2, 100172, 2024. doi:10.1016/j.tbench.2024.100172